20261020 Quiz

for (MATH301) Modern Algebra I

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Exercise 3.1.14. Consider the additive quotient group $\mathbb{Q}/\mathbb{Z}$.

(a) Show that every coset of $\mathbb{Z}$ in $\mathbb{Q}$ contains exactly one representative $q \in \mathbb{Q}$ with $0 \leq q < 1$.

(b) Show that every element of $\mathbb{Q}/\mathbb{Z}$ has finite order, but that elements of arbitrarily large order exist.

(c) Show that $\mathbb{Q}/\mathbb{Z}$ is the torsion subgroup of $\mathbb{R}/\mathbb{Z}$ (cf. Exercise 2.1.6).

(d) Prove that $\mathbb{Q}/\mathbb{Z}$ is isomorphic to the multiplicative group of roots of unity in $\mathbb{C}^\times$.

Solution.

(a) Given $q \in \mathbb{Q}$, let $m = \lfloor q \rfloor$. Then $q - m \in [0,1) \cap \mathbb{Q}$ and $(q-m) + \mathbb{Z} = q + \mathbb{Z}$. If $u, v \in [0,1)$ represent the same coset, then $u - v \in \mathbb{Z} \cap (-1,1) = \{0\}$, so $u = v$.

(b) Write $q = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$. Then $b(q + \mathbb{Z}) = \mathbb{Z}$, so every element has finite order. Moreover, $1/n + \mathbb{Z}$ has order n for each $n \geq 1$.

(c) For $x \in \mathbb{R}$, the coset $x + \mathbb{Z}$ has finite order if and only if $nx \in \mathbb{Z}$ for some positive integer n, equivalently $x \in \mathbb{Q}$. Thus the torsion subgroup is precisely $\mathbb{Q}/\mathbb{Z}$.

(d) Define

$$\varphi \colon \mathbb{Q}/\mathbb{Z} \to \mathbb{C}^\times, \qquad q + \mathbb{Z} \mapsto e^{2\pi i q}.$$

This is a well-defined homomorphism because adding an integer to q does not change its image and exponentials turn sums into products. Its kernel is trivial since $e^{2\pi i q} = 1$ exactly when $q \in \mathbb{Z}$. Every root of unity z satisfies $z^n = 1$ for some $n > 0$, hence $z = e^{2\pi i k/n}$ for some $k \in \mathbb{Z}$. Therefore φ is an isomorphism onto the group of roots of unity.

□

Exercise 3.1.17. Let $G = D_8$ be the dihedral group of order 16 (whose subgroup lattice appears in §2.5), with generators r, s satisfying

$$r^8 = s^2 = 1, \qquad rs = sr^{-1},$$

The subgroup $\langle r^4 \rangle = Z(G)$ is normal. Let $\overline{G} = G/\langle r^4 \rangle$, with $\bar{r} = r\langle r^4 \rangle$ and $\bar{s} = s\langle r^4 \rangle$.

(a) Show that $|\overline{G}| = 8$.

(b) Exhibit every element of $\overline{G}$ in the form $\bar{s}^a \bar{r}^b$ for integers a, b.

(c) Find the order of each element listed in part (b).

(d) Express $\overline{rs}$, $\overline{sr^{-2}s}$, and $\overline{s^{-1}r^{-1}sr}$ in the form $\bar{s}^a \bar{r}^b$.

(e) Prove that $\overline{H} = \langle \bar{s}, \bar{r}^2 \rangle$ is normal in $\overline{G}$ and isomorphic to V_4. Determine the isomorphism type of its complete preimage in G.

(f) Find $Z(\overline{G})$ and determine the isomorphism type of $\overline{G}/Z(\overline{G})$.

Notation. Following MATH000, D_n is the symmetry group of a regular n-gon and has order $2n$. Thus D_8 here is denoted by D_{16} in Dummit–Foote.

Solution. Since r^4 commutes with both r and s, the subgroup $\langle r^4 \rangle$ is central and hence normal.

(a) $|\langle r^4 \rangle| = 2$, so $|\overline{G}| = 16/2 = 8$.

(b) The eight distinct elements are

$$1, \bar{r}, \bar{r}^2, \bar{r}^3, \bar{s}, \bar{s}\bar{r}, \bar{s}\bar{r}^2, \bar{s}\bar{r}^3.$$

Indeed, $\bar{r}^4 = \bar{s}^2 = 1$ and $\bar{r}\bar{s} = \bar{s}\bar{r}^{-1}$.

(c) Their orders, in the order displayed in (b), are $1, 4, 2, 4, 2, 2, 2, 2$, respectively.

(d) Using $r^j s = sr^{-j}$ and $s^{-1} = s$,

$$\overline{rs} = \bar{s}\bar{r}^3, \qquad \overline{sr^{-2}s} = \bar{r}^2, \qquad \overline{s^{-1}r^{-1}sr} = \bar{r}^2.$$

(e) The subgroup

$$\overline{H} = \{1, \bar{r}^2, \bar{s}, \bar{s}\bar{r}^2\}$$

has order 4, hence index 2 in $\overline{G}$ and is normal. Its three nonidentity elements have order 2, so $\overline{H} \cong V_4$. Its complete preimage is $\langle r^2, s \rangle$, where r^2 has order 4 and $s(r^2)s = (r^2)^{-1}$. This subgroup has order 8 and is isomorphic to D_4.

(f) The central elements are 1 and $\bar{r}^2$, so $Z(\overline{G}) = \langle \bar{r}^2 \rangle$. The quotient has four elements, all of whose nonidentity elements have order 2, hence $\overline{G}/Z(\overline{G}) \cong V_4$.

□

Exercise 3.1.22.

(a) Prove that if H and K are normal subgroups of a group G, then $H \cap K \trianglelefteq G$.

(b) Prove that the intersection of an arbitrary nonempty collection of normal subgroups of a group is normal (the collection need not be countable).

Solution.

(a) The intersection $H \cap K$ is a subgroup. For each $g \in G$,

$$g(H \cap K)g^{-1} = (gHg^{-1}) \cap (gKg^{-1}) = H \cap K.$$

Thus $H \cap K \trianglelefteq G$.

(b) Let $(N_i)_{i \in I}$ be a nonempty collection with $N_i \trianglelefteq G$. Their intersection N is a subgroup, and for any $g \in G$,

$$gNg^{-1} = \bigcap_{i \in I} gN_ig^{-1} = \bigcap_{i \in I} N_i = N.$$

Therefore $N \trianglelefteq G$.

□

Exercise 3.1.24. Prove that if $N \trianglelefteq G$ and $H \leq G$, then $N \cap H \trianglelefteq H$.

Solution. The intersection $N \cap H$ is a subgroup of H. For $h \in H$ and $x \in N \cap H$, normality of N gives $hxh^{-1} \in N$, while $h, x \in H$ gives $hxh^{-1} \in H$. Hence $h(N \cap H)h^{-1} \subseteq N \cap H$, and the reverse inclusion follows by replacing h with h^{-1}. Thus $N \cap H \trianglelefteq H$.

□

Exercise 3.1.31. Prove that if $H \leq G$ and $N \trianglelefteq H$, then $H \leq N_G(N)$. Deduce that $N_G(N)$ is the largest subgroup of G in which N is normal, i.e., the join of all subgroups H for which $N \trianglelefteq H$.

Solution. If $h \in H$, then $hNh^{-1} = N$ because $N \trianglelefteq H$. Thus $h \in N_G(N)$ and $H \leq N_G(N)$. Conversely, $N \leq N_G(N)$ and $N \trianglelefteq N_G(N)$ by the definition of the normalizer. Therefore $N_G(N)$ itself belongs to the stated family and contains every other member. It is consequently its largest member and its join.

□

Exercise 3.1.33. Find all normal subgroups of D_4 and determine the isomorphism type of each corresponding quotient.

Hint. You may use the subgroup lattice for D_4 in §2.5.

Solution. Write $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$. Its three subgroups of index 2 are normal, as are $\{1\}$, $\langle r^2 \rangle = Z(D_4)$, and D_4. These are all: conjugating a reflection by r changes it by a factor r^2, so no reflection subgroup of order 2 is normal. The normal subgroups and quotients are

N	D_4/N
$\{1\}$	D_4
$\langle r^2 \rangle$	V_4
$\langle r \rangle$	C_2
$\langle r^2, s \rangle$	C_2
$\langle r^2, sr \rangle$	C_2
D_4	$\{1\}$

For the central quotient, the images of r and s commute and both have order 2, giving V_4; the index-2 quotients have order 2.

□

Exercise 3.1.34. Let $n \geq 3$ and let

$$D_n = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle.$$

For each positive divisor k of n:

(a) Prove that $\langle r^k \rangle \trianglelefteq D_n$.

(b) Prove that $D_n/\langle r^k \rangle$ is a dihedral group of order $2k$.

Solution.

(a) The subgroup $\langle r^k \rangle$ is preserved by conjugation by r and by s, since

$$r(r^k)r^{-1} = r^k, \qquad s(r^k)s^{-1} = r^{-k}.$$

As r, s generate D_n, it is normal.

(b) In the quotient let $\bar{r} = r\langle r^k \rangle$ and $\bar{s} = s\langle r^k \rangle$. We have

$$\bar{r}^k = \bar{s}^2 = 1, \qquad \bar{s}\bar{r}\bar{s} = \bar{r}^{-1}.$$

Since $k \mid n$, the subgroup $\langle r^k \rangle$ has order n/k and the quotient has order $2k$. The elements $\bar{r}^i$ and $\bar{s}\bar{r}^i$ $(0 \leq i < k)$ are distinct, giving the dihedral presentation of order $2k$. For $k \geq 3$ this is D_k in MATH000 notation; the degenerate cases $k = 1, 2$ yield C_2 and V_4, respectively.

□

Exercise 3.1.40. Let G be a group, let $N \trianglelefteq G$, and write $\overline{G} = G/N$. Prove that $\bar{x}$ and $\bar{y}$ commute in $\overline{G}$ if and only if $[x, y] \in N$, where $[x, y] = x^{-1}y^{-1}xy$ is the commutator of x and y.

Solution. The elements $\bar{x}$ and $\bar{y}$ commute if and only if

$$(\bar{x})^{-1}(\bar{y})^{-1}\bar{x}\bar{y} = 1_{\overline{G}}.$$

The left-hand side is $\overline{x^{-1}y^{-1}xy} = \overline{[x, y]}$, which equals the identity if and only if $[x, y] \in N$.

□

Exercise 3.1.41. Let G be a group. Prove that $N = \langle [x, y] \mid x, y \in G \rangle$ is a normal subgroup of G and that G/N is abelian. The subgroup N is called the commutator subgroup of G.

Solution. Conjugation preserves commutators: for $g, x, y \in G$,

$$g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}].$$

Thus $gNg^{-1} = N$ for every $g \in G$, so $N \trianglelefteq G$. Every commutator lies in N, hence by Exercise 3.1.40 the images of any $x, y \in G$ commute in G/N. Therefore G/N is abelian. In MATH000 notation, $N = [G, G]$.

□

Exercise 3.2.4. Show that if $|G| = pq$ for primes p, q (not necessarily distinct), then either G is abelian or $Z(G) = \{1\}$.

Hint. See Exercise 3.1.36.

Solution. Suppose G is not abelian. If $Z(G) \neq \{1\}$, then $Z(G)$ is a proper nontrivial subgroup, so Lagrange's Theorem gives $|Z(G)| = p$ or q (when such a proper subgroup exists). Consequently $G/Z(G)$ has prime order and is cyclic. But a group whose quotient by its center is cyclic is abelian: if $G/Z(G) = \langle xZ(G) \rangle$, every pair of elements can be written as $x^a z$ and $x^b w$ with $z, w \in Z(G)$, and such elements commute. This contradicts the assumption. Hence $Z(G) = \{1\}$.

□

Exercise 3.2.5. Let $H \leq G$ and fix $g \in G$.

(a) Prove that gHg^{-1} is a subgroup of G of the same order as H.

(b) Deduce that if $n \geq 1$ and H is the unique subgroup of G of order n, then $H \trianglelefteq G$.

Solution.

(a) Conjugation $c_g \colon G \to G$, $h \mapsto ghg^{-1}$, is an automorphism. Hence its image gHg^{-1} is a subgroup, and its restriction to H is a bijection onto that subgroup. Thus $|gHg^{-1}| = |H|$, including when H is infinite.

(b) For every $g \in G$, the subgroup gHg^{-1} has order n. By uniqueness, $gHg^{-1} = H$, which is exactly $H \trianglelefteq G$.

□

Exercise 3.2.8. Prove that if H and K are finite subgroups of G of relatively prime orders, then $H \cap K = \{1_G\}$.

Solution. Since $H \cap K$ is a subgroup of both H and K, Lagrange's Theorem implies that $|H \cap K|$ divides both $|H|$ and $|K|$. Their greatest common divisor is 1, so $|H \cap K| = 1$ and $H \cap K = \{1_G\}$.

□

Exercise 3.2.11. Let $H \leq K \leq G$. Prove that

$$[G : H] = [G : K][K : H],$$

without assuming that G is finite.

Solution. Choose representatives $(g_i)_{i\in I}$ for the left cosets of K in G, and representatives $(k_j)_{j\in J}$ for the left cosets of H in K. Every $g \in G$ has a unique expression at the coset level $g \in g_i k_j H$: first $g \in g_i K$, then $g_i^{-1} g \in k_j H$. Moreover, $g_i k_j H = g_{i'} k_{j'} H$ implies $g_i K = g_{i'} K$, so $i = i'$, and then $k_j H = k_{j'} H$, so $j = j'$. Hence the cosets of H in G are in bijection with $I \times J$, giving $[G : H] = [G : K][K : H]$ (as a cardinal equality, even for infinite indices).

□

Exercise 3.2.14. Prove that S_4 has neither a normal subgroup of order 8 nor a normal subgroup of order 3.

Solution. Suppose $N \trianglelefteq S_4$ and $|N| = 8$. Then S_4/N has order 3, so the quotient homomorphism sends every transposition to the identity, since its image has order dividing both 2 and 3. Every cycle $(a_1\ a_2\ \cdots\ a_k)$ is a product of transpositions, namely $(a_1\ a_k) \cdots (a_1\ a_2)$. Thus transpositions generate S_4, and the quotient map would be trivial, contradicting $|S_4/N| = 3$.

If instead $|N| = 3$, then N contains a 3-cycle. All 3-cycles in S_4 are conjugate, so normality forces all eight 3-cycles to lie in N. Together with the identity this gives at least nine elements, a contradiction.

□

Exercise 3.2.16. Use Lagrange's Theorem in $(\mathbb{Z}/p\mathbb{Z})^\times$ to prove Fermat's Little Theorem: if p is prime, then

$$a^p \equiv a \pmod{p} \qquad \text{for every } a \in \mathbb{Z}.$$

Solution. If $p \mid a$, both sides are 0 modulo p. Otherwise $[a]_p$ belongs to the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$, which has order $p - 1$. By Lagrange's Theorem,

$$[a]_p^{p-1} = [1]_p,$$

so $a^{p-1} \equiv 1 \pmod{p}$. Multiplying by a gives $a^p \equiv a \pmod{p}$.

□

Exercise 3.2.20. If A is an abelian normal subgroup of G and B is any subgroup of G, prove that $A \cap B \trianglelefteq AB$.

Solution. Since $A \trianglelefteq G$, the product AB is a subgroup of G. Set $N = A \cap B$. For $b \in B$ and $x \in N$, normality of A gives $bxb^{-1} \in A$, and subgroup closure gives $bxb^{-1} \in B$. Hence B normalizes N. Since A is abelian, each $a \in A$ centralizes N. Thus both A and B normalize N, so their product AB normalizes N. As $N \leq AB$, this proves $N \trianglelefteq AB$.

□

Exercise 3.2.21. Prove that $\mathbb{Q}$ has no proper subgroups of finite index. Deduce that $\mathbb{Q}/\mathbb{Z}$ has no proper subgroups of finite index.

Hint. Recall Exercises 1.6.21 and 3.1.15.

Solution. Let $H \leq (\mathbb{Q}, +)$ have finite index m. Since $\mathbb{Q}$ is abelian, $\mathbb{Q}/H$ is a group of order m. For any $q \in \mathbb{Q}$, Lagrange's Theorem gives $mq \in H$. As $q = m(q/m)$, applying this to $q/m \in \mathbb{Q}$ shows $q \in H$. Thus $H = \mathbb{Q}$.

If $K \leq \mathbb{Q}/\mathbb{Z}$ has finite index, its preimage under the natural projection $\pi\colon \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ has the same finite index in $\mathbb{Q}$. By the first part, $\pi^{-1}(K) = \mathbb{Q}$, so $K = \mathbb{Q}/\mathbb{Z}$.

□

Exercise 3.3.3. Prove that if $H \trianglelefteq G$ has prime index p, then for every $K \leq G$ either

(i) $K \leq H$, or

(ii) $G = HK$ and $[K : K \cap H] = p$.

Solution. Since $H \trianglelefteq G$, the product HK is a subgroup containing H. As $[G : H] = p$ is prime, the index formula forces $HK = H$ or $HK = G$. In the first case $K \leq H$. In the second, the Second Isomorphism Theorem gives

$$K/(K \cap H) \cong HK/H = G/H,$$

so $[K : K \cap H] = |G/H| = p$.

□

Exercise 3.3.4. Let $C \trianglelefteq A$ and $D \trianglelefteq B$. Prove that

$$C \times D \trianglelefteq A \times B \quad \text{and} \quad (A \times B)/(C \times D) \cong (A/C) \times (B/D).$$

Solution. For $(a, b) \in A \times B$,

$$(a, b)(C \times D)(a, b)^{-1} = (aCa^{-1}) \times (bDb^{-1}) = C \times D,$$

so $C \times D \trianglelefteq A \times B$. Define

$$\varphi\colon A \times B \to (A/C) \times (B/D), \qquad (a, b) \mapsto (aC, bD).$$

This is a surjective homomorphism with kernel $C \times D$. The First Isomorphism Theorem yields the claimed isomorphism.

□

Exercise 3.5.3. Prove that S_n is generated by the adjacent transpositions

$$\{(i\ i+1) \mid 1 \leq i \leq n-1\}.$$

Hint. Consider conjugates, for example $(2\ 3)(1\ 2)(2\ 3)^{-1}$.

Solution. Let $s_i = (i\ i+1)$. For $1 \leq i < j \leq n$,

$$(i\ j) = s_i s_{i+1} \cdots s_{j-2} s_{j-1} s_{j-2} \cdots s_{i+1} s_i.$$

Thus every transposition is generated by adjacent transpositions. Since every permutation is a product of transpositions, these elements generate S_n.

□

Exercise 3.5.4. Show that $S_n = \langle (12), (1\ 2\ \cdots\ n) \rangle$ for all $n \geq 2$.

Solution. Set $s = (12)$ and $c = (1\ 2\ \cdots\ n)$. For $1 \le i \le n-1$, conjugation gives

$$c^{i-1} s c^{-(i-1)} = (i\ i+1).$$

Hence $\langle s, c\rangle$ contains all adjacent transpositions. By Exercise 3.5.3 these generate S_n, so $\langle s, c\rangle = S_n$.

□

Exercise 5.1.1. Show that the center of a direct product is the direct product of its centers:

$$Z(G_1 \times \cdots \times G_n) = Z(G_1) \times \cdots \times Z(G_n).$$

Deduce that a direct product of groups is abelian if and only if each factor is abelian.

Solution. An element $(g_1, \ldots, g_n)$ commutes with every $(h_1, \ldots, h_n)$ if and only if $g_i h_i = h_i g_i$ for every i and every $h_i \in G_i$. This is equivalent to $g_i \in Z(G_i)$ for all i, proving the identity. The product is abelian precisely when its center is the whole product, equivalently $Z(G_i) = G_i$ for every factor.

□

Exercise 5.1.14. Let $G = A_1 \times \cdots \times A_n$, and for each i let $B_i \trianglelefteq A_i$. Prove that

$$B_1 \times \cdots \times B_n \trianglelefteq G$$

and

$$\frac{A_1 \times \cdots \times A_n}{B_1 \times \cdots \times B_n} \cong (A_1/B_1) \times \cdots \times (A_n/B_n).$$

Solution. Conjugation in G acts coordinatewise, so for $(a_i) \in G$,

$$(a_i)(B_1 \times \cdots \times B_n)(a_i)^{-1} = (a_1 B_1 a_1^{-1}) \times \cdots \times (a_n B_n a_n^{-1}) = B_1 \times \cdots \times B_n.$$

Define the coordinatewise quotient homomorphism

$$\varphi \colon G \to \prod_{i=1}^{n} (A_i/B_i), \qquad (a_1, \ldots, a_n) \mapsto (a_1 B_1, \ldots, a_n B_n).$$

It is surjective and has kernel $\prod_{i=1}^{n} B_i$. The result follows from the First Isomorphism Theorem.

□

Exercise 5.2.2. For each of the following orders, give the lists of invariant factors for all abelian groups of that order:

(a) 270,

(b) 9801,

(c) 320,

(d) 105,

(e) 44100.

Solution. We write an invariant-factor list as $(n_1, \ldots, n_t)$, with $n_{i+1} \mid n_i$, corresponding to $C_{n_1} \times \cdots \times C_{n_t}$. The Fundamental Theorem of Finite Abelian Groups gives one list for each choice of partitions of the exponents in the prime factorization. The complete lists are:

(a) 270:

$$(270) \qquad (90, 3) \qquad (30, 3, 3)$$

(b) 9801:

$$\begin{array}{ccc} (9801) & (3267,3) & (1089,9) \\ (1089,3,3) & (891,11) & (363,3,3,3) \\ (297,33) & (99,99) & (99,33,3) \\ & (33,33,3,3) & \end{array}$$

(c) 320:

$$\begin{array}{ccc} (320) & (160,2) & (80,4) \\ (80,2,2) & (40,8) & (40,4,2) \\ (40,2,2,2) & (20,4,4) & (20,4,2,2) \\ (20,2,2,2,2) & (10,2,2,2,2,2) & \end{array}$$

(d) 105:

$$(105)$$

(e) 44100:

$$\begin{array}{ccc} (44100) & (22050,2) & (14700,3) \\ (8820,5) & (7350,6) & (6300,7) \\ (4410,10) & (3150,14) & (2940,15) \\ (2100,21) & (1470,30) & (1260,35) \\ (1050,42) & (630,70) & (420,105) \\ & (210,210) & \end{array}$$

There are respectively $3, 10, 11, 1$, and 16 lists, in agreement with the products of the partition numbers for the prime-power exponents.

□

Exercise 5.2.5. Let G be a finite abelian group of type $(n_1, n_2, \ldots, n_r)$. Prove that G contains an element of order m if and only if $m \mid n_1$. Deduce that G is of exponent n_1.

Solution. In invariant-factor form,

$$G \cong C_{n_1} \times \cdots \times C_{n_r}, \qquad n_r \mid n_{r-1} \mid \cdots \mid n_1.$$

The order of any element $(x_1, \ldots, x_r)$ is the least common multiple of the orders of its coordinates, so it divides n_1. Conversely, if $m \mid n_1$, the first factor C_{n_1} has an element of order m; inserting this element in the first coordinate and identities elsewhere gives an element of order m in G. The largest element order and the least common multiple of all element orders are consequently n_1, which is the exponent of G.

□

Exercise 5.2.9. Let $A = C_{60} \times C_{45} \times C_{12} \times C_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in A.

Solution. Exactly three factors have even order: C_{60}, C_{12}, C_{36}. Each has exactly two solutions of $x^2 = 1$, while C_{45} has only the identity. Thus there are $2^3 = 8$ elements satisfying $x^2 = 1$, of which one is the identity. The number of elements of order 2 is therefore $\boxed{7}$.

Every index-2 subgroup is the kernel of a surjective homomorphism $A \to C_2$. A homomorphism to C_2 has two choices on each even-order cyclic factor and only the trivial choice on the odd-order factor. Hence there are $2^3 - 1 = 7$ nontrivial homomorphisms. They are surjective, and distinct ones have distinct kernels because $\mathrm{Aut}(C_2)$ is trivial. Thus the number of index-2 subgroups is also $\boxed{7}$.

□